How-To Library

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Converting Hierarchical Input to Flat Output using Microsoft Azure Cosmos DB SQL API Connector

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Abstract

When you use Microsoft Azure Cosmos DB SQL API Connector to perform extensive query operations on data, you can use the Hierarchical Parser transformation to convert hierarchical input into flat output. The transformation processes XML or JSON input from the upstream transformation and provides flat output to the downstream transformation. This article describes how to use the Hierarchical Parser transformation with Microsoft Azure Cosmos DB SQL API Connector to convert hierarchical input into flat output.

Supported Versions

- Cloud Data Integration

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Overview

You can use the Hierarchical Parser transformation with Microsoft Azure Cosmos DB SQL API Connector to convert hierarchical input into flat output. Use a schema file to create a hierarchical schema that defines the expected hierarchy of the input data. Create and configure the Hierarchical Parser transformation in Data Integration and add it to a mapping. When you run the mapping, the Hierarchical Parser transformation converts the hierarchical input based on the schema file and converts the hierarchical input into the flat output.

To configure a Hierarchical Parser transformation in a Microsoft Azure Cosmos DB SQL API mapping, perform the following steps:

1. Create a hierarchical schema.
2. Add a Source transformation to the mapping.
3. Add a Hierarchical Parser transformation to the mapping.
    a. Associate the hierarchical schema with the Hierarchical Parser transformation.
    b. Configure the field mapping to select the schema elements that provide flat output.
4. Add a Target transformation to the mapping.

Before You Begin

Verify that you have the permission to use the following licenses and packages before you create a mapping:

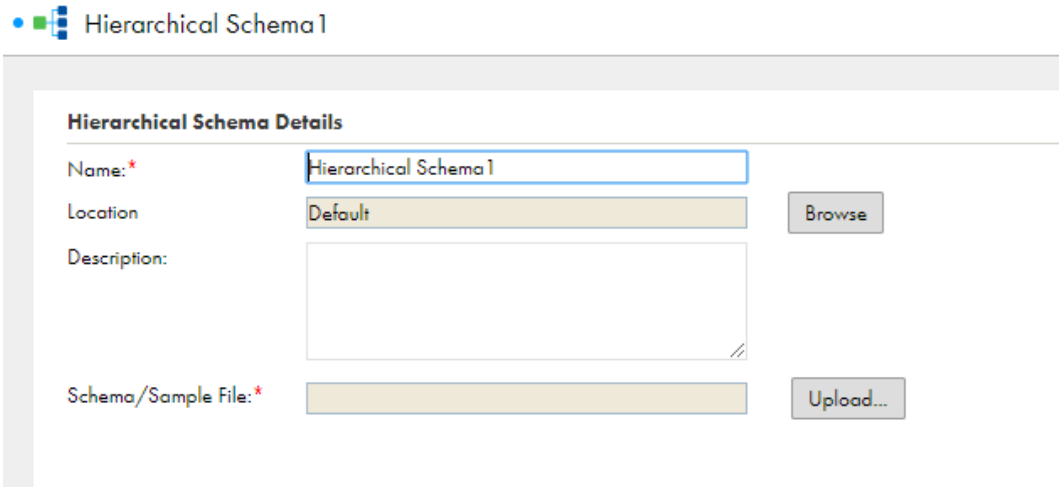
- Agent Compiler
- DT XML Processing (License)
- Data Transformation
- saas-xmetadataread
- UTDforHierarchy

Create a Hierarchical Schema

A hierarchical schema is based on a schema file that you import into Data Integration. The schema defines the expected hierarchy of the input data. After you import a schema file for Microsoft Azure Cosmos DB SQL API Connector, select the schema root.

1. In Data Integration, click **New** > **Components** > **Hierarchical Schema**.

   The **Hierarchical Schema 1** page appears.



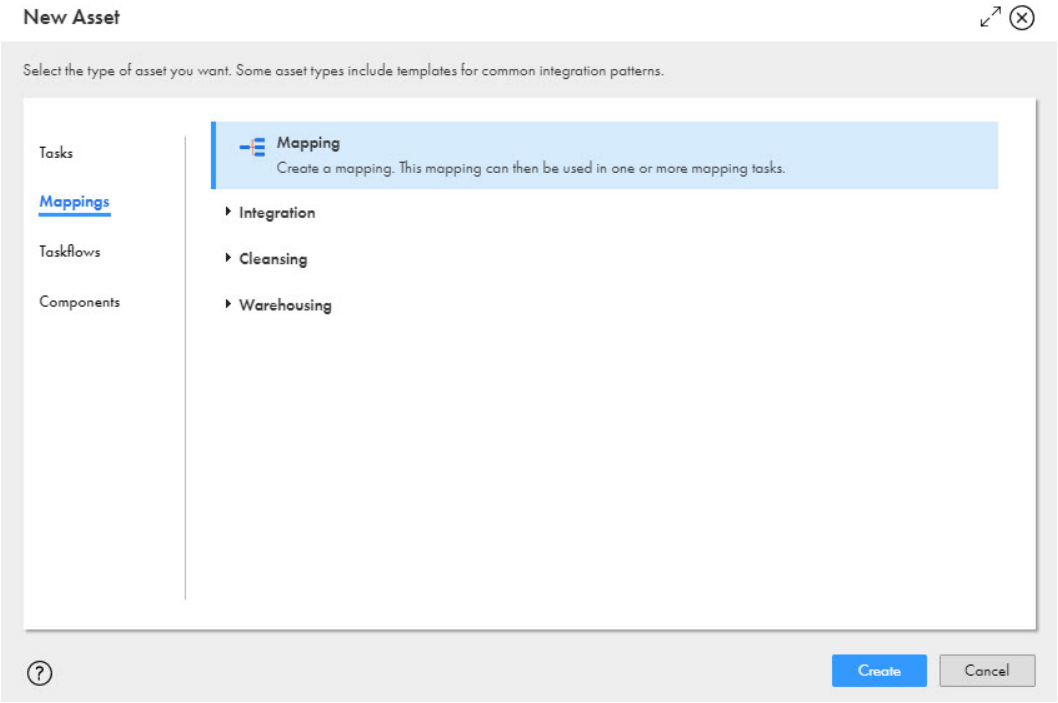
2. Enter a name for the hierarchical schema and optionally enter a description.
3. To select a schema, click **Upload**, browse for a schema file, and click **OK**.
4. To save the hierarchical schema, click **Save**.

Add a Source Transformation to the Mapping

Add a source transformation to a mapping to specify the Microsoft Azure Cosmos DB SQL API source object.

1. To create a mapping, click **New** and then in the **New Asset** dialog box, click **Mapping**.

   The following image shows the **New Asset** dialog box with **Mapping** selected:



2. Click **Create**. The **Mapping Designer** appears with a new mapping displayed in the mapping canvas.
3. On the mapping canvas, click the Source transformation.
4. Specify the name and description of the source object.
5. Click **Source** and select an active Microsoft Azure Cosmos DB SQL API connection.
6. Select `Single Object` as Source Type.
7. Select a Cosmos DB collection as the Microsoft Azure Cosmos DB SQL API object from where you want to read the hierarchical data.
8. Select `None` as **Format Type** in **Formatting Options**.

   When you select the format type as None, the data is read as binary.
9. Configure advance properties for the source object as required.

Add a Hierarchical Parser Transformation to the Mapping

Add a Hierarchy Parser transformation to the mapping and connect it to the upstream Source transformation. Configure the transformation properties.

Input Setting

Select the hierarchical schema and input type on the **Input Settings** tab of the **Properties** panel. The input type can be a buffer or a file.

1. Click the **Input Settings** tab under the **Properties** panel.

   The following image shows the input settings properties:



2. Select **Buffer** or **File** as an **Input Type**.
3. Click **Select**.

   The **Select Schema** dialog box appears.
4. Choose the hierarchical schema from the list to include in the Hierarchical Parser transformation and click **Select**.

Input Field Selection

Configure the input field selection in a Hierarchy Parser transformation to define which field in the upstream transformation to map to the input field in the Hierarchy Parser transformation.

Configure the Input Field Selection tab of the Properties panel.

You can configure the following input field selection options:

Automatch

Links fields with matching names. To map a field, use the Map Selected option from the actions near the Automatch drop-down list. After you map an incoming field to the hierarchical schema input field, to change the mapping you must first clear the mapping. Use the Clear Mapping or Unmap Selected option.

Options

Controls the fields that appear in the Incoming Fields list. Shows all fields, unmapped fields, or mapped fields. Determines how field names appear in the input field list. Use technical field names or labels.

Configure the Field Mapping to Select Schema Elements

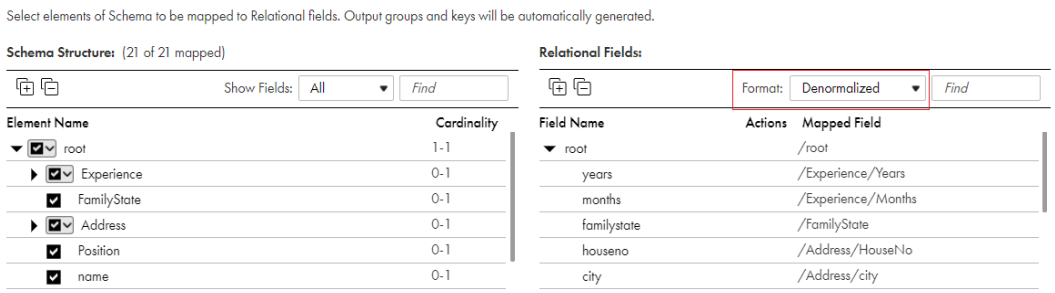
To define the schema elements that provide the relational output, configure the field mapping in a Hierarchical Parser transformation.

1. In the **Properties** panel of the Hierarchical Parser transformation, click the **Field Mapping** tab.

   The field mapping editor displays the transformation input elements and output fields. To the left, the editor shows the schema elements. To the right, the editor shows the output fields.

2. Configure the field mapping to select which schema elements provide flat output.

   The following image shows the field mapping properties:



3. To view child elements, expand an element.
4. To search for an element, enter the search characters in the **Search** field, and press **Enter**.
5. To denormalize the output, select **Denormalized** in the **Format** field, and then select the elements that you want to include in the output.

   By default, the output is not denormalized, and the setting for the **Format** field is **Relational**.
6. To include an element without child elements in the output, select the element.
7. To exclude an element without child elements from the output, clear the element.
8. For an element with child elements, you can choose from the following options:
   a. To include all child elements in the output, select the check box against the root element and select **Map all descendants**.
   b. To include immediate child elements in the output, right-click the element and select **Map immediate children**.
   c. To exclude all child elements in the output, right-click the element and select **Unmap all descendants**.
   d. To exclude immediate child elements in the output, right-click the element and select **Unmap immediate children**.
9. To search for a field, enter the search characters the **Search** field, and press **Enter**.
10. To exclude a relational output field from the output, clear the check box against the element name.

Output Fields

When you select schema elements to use in the Hierarchical Parser transformation, the schema output fields appear on the **Output Fields** tab of the **Properties** panel.

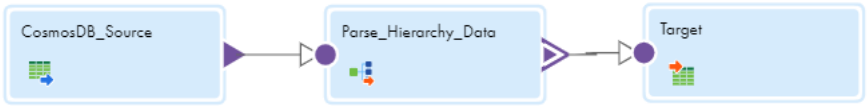
The output editor displays the name, type, precision, scale, and origin for each output field in each output group. To edit the name and precision values in the output field, click the name and the precision elements to edit the value.

Add a Target Transformation to the Mapping

Add a target transformation to the mapping and connect it to the upstream Hierarchy Parser transformation.

1. Select a flat file as a target object.
2. Configure incoming field rules and output fields for the transformation.

   The following image shows the mapping:



3. Save and run the mapping.

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